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Is the CS2 Skin Market Integrated with Traditional and Crypto Financial Markets? A Time Series Analysis

Time Series Analysis in Finance (TSA01_FS2601)

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Abstract

This paper examines whether the CS2 skin market — a multi-billion-dollar virtual goods market operating through the Steam Marketplace — is integrated with traditional and crypto financial markets. We construct a daily value-weighted price index from 1,607 liquid CS2 skins over the period August 2013 to June 2024 and test for market integration against the S&P 500, Bitcoin, and Gold using exploratory data analysis, GARCH(1,1) volatility modelling, and a VAR(4) model with Granger causality tests. We find no evidence of market integration: contemporaneous correlations between CS2 returns and all three comparison assets are indistinguishable from zero, and no Granger causal relationship is detected in either direction at the 5% significance level. CS2 exhibits extreme annualised volatility (114.56%) and GARCH persistence identical to Bitcoin (0.9990), but its price dynamics are driven by game-specific factors rather than macroeconomic forces. These results position CS2 skins as a genuinely segmented alternative asset class with potential diversification benefits for financial investors.

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List of Abbreviations

ACF	Autocorrelation Function
ADF	Augmented Dickey-Fuller (unit root test)
AIC	Akaike Information Criterion
BIC	Bayesian Information Criterion (Schwarz Criterion, SC)
CS2	Counter-Strike 2
FPE	Final Prediction Error
GARCH	Generalised Autoregressive Conditional Heteroskedasticity
HQ	Hannan-Quinn Information Criterion
PACF	Partial Autocorrelation Function
VAR	Vector Autoregression
VWAP	Volume-Weighted Average Price

Glossary

CS2 (Counter-Strike 2) Valve's tactical first-person shooter and the successor to Counter-Strike: Global Offensive (CS:GO). Launched in 2023, it operates the same Steam Marketplace ecosystem established in 2013, through which cosmetic items are traded at real-money prices.

Skin A purely cosmetic weapon or equipment modification in CS2 with no gameplay effect. Skins are bought, sold, and stored as digital assets on the Steam Marketplace and constitute the asset class studied in this paper.

StatTrak A skin variant equipped with an electronic counter that tracks the number of in-game kills. StatTrak skins command a persistent price premium relative to their non-StatTrak counterparts and are excluded from the index to avoid confounding price signals unrelated to market-wide demand.

Steam Marketplace Valve Corporation's open secondary market integrated into the Steam gaming platform, through which users buy and sell in-game items at prices set by supply and demand. Unlike closed in-game economies, the Steam Marketplace allows freely negotiated prices denominated in real currency, making it amenable to standard financial time series analysis.

Value-weighted index A price index in which each constituent's contribution to the index level is proportional to its trading volume. Formally, $I_t = \sum_i p_{it} v_{it} / \sum_i v_{it}$, where p_{it} is the volume-weighted average price of item i on day t and v_{it} is its trading volume. Analogous to market-capitalisation weighting in equity indices, this ensures that heavily traded items drive the index level.

Log return The continuously compounded return defined as $r_t = \ln(P_t/P_{t-1})$. Preferred over simple returns because log returns are approximately additive over time, yield a stationary series when prices are non-stationary, and are symmetric around zero for up and down moves of equal magnitude.

Granger causality A statistical concept introduced by Granger Granger (1969): variable X Granger-causes Y if past values of X improve forecasts of Y beyond what is achievable using Y's own history alone. Granger causality is a statement about predictability, not structural or physical causation.

Liquidity filter A selection criterion that excludes assets with insufficient trading activity from an index. In this paper, any base item recording fewer than two transactions in any calendar month is excluded, analogous to the SIX Swiss Exchange minimum turnover requirement for equity index inclusion SIX Swiss Exchange AG (2024).

Persistence ($\hat{\alpha} + \hat{\beta}$) In a GARCH(1,1) model, the sum of the ARCH coefficient $\hat{\alpha}$ and the GARCH coefficient $\hat{\beta}$. Values close to unity indicate slow mean-reversion of conditional variance: a volatility shock today affects expected variance far into the future. A value of exactly 1 defines an integrated GARCH (IGARCH) process.

1 Introduction

Counter-Strike 2 (CS2) is one of the most played competitive shooters in the world. Since 2013, players have been able to trade purely cosmetic weapon modifications — "skins" — through the Steam Marketplace at freely negotiated, real-money prices. Unlike the closed monetisation systems of competing titles, Steam operates an open secondary market, making CS2 skins an observable and quantifiable asset class. By early 2025, the ecosystem had grown to over 22,000 tradeable items with an estimated market value of $4.1 billion Guede-Fernández et al. (2025). Prior work has found attractive risk-adjusted returns and low correlation with traditional assets Dobrynskaya and Strelnikov (2025), but no study has formally tested market integration using a VAR and Granger causality framework on a comprehensively constructed price index.

This paper fills that gap. We construct a daily value-weighted CS2 price index from raw Steam Marketplace transaction data and ask:

"Is the CS2 skin market integrated with traditional and crypto financial markets, and what is the direction and temporal dynamics of any such relationship?"

We test this against the S&P 500, Bitcoin, and Gold — benchmarks for traditional equity, speculative crypto, and safe-haven performance — using EDA, GARCH(1,1) volatility modelling, and a VAR model with Granger causality tests.

2 Literature Review

The CS2 skin market constitutes a substantial financial asset class. Operating through the Steam Marketplace — an open secondary market with real-currency pricing — it reached an estimated $4.1 billion in value by early 2025 Guede-Fernández et al. (2025). Unlike competing titles that restrict trading to developer-set prices, CS2's open market allows supply and demand to determine prices, making skins suitable for standard financial analysis Zhou (2024). Dobrynskaya and Strelnikov document average annual returns of 41.2% with a Sharpe ratio of 0.34 across nearly 5,000 CS skins from 2013 to 2024 — exceeding both the US equity market (0.25) and gold (0.12) Dobrynskaya and Strelnikov (2025). These findings motivate the central question of whether the CS2 market operates independently of, or co-moves with, mainstream financial assets.

Bitcoin serves as the primary speculative comparator. Baur et al. find that Bitcoin is held predominantly for speculative purposes, with returns uncorrelated with traditional asset classes under both normal and stressed market conditions Baur et al. (2018). Bouri et al. confirm that Bitcoin offers minimal hedging properties, with its portfolio value lying in diversification rather than risk reduction Bouri et al. (2017). The shared demographic between Bitcoin investors and CS2 skin traders — young, digitally native, and risk-tolerant — motivates the hypothesis that crypto market sentiment may Granger-cause skin prices. Akin et al., using a VAR framework equivalent to the one applied here, find weak but statistically detectable interactions between Bitcoin, the S&P 500, and gold, establishing the empirical context against which CS2's behaviour is measured Akin et al. (2024).

The methodological toolkit draws on three established approaches. Vector Autoregression, introduced by Sims, captures joint dynamics across multiple time series without imposing a-priori causal structure Sims (1980). Granger causality Granger (1969) operationalises directional predictability: X Granger-causes Y if past values of X improve forecasts of

Y beyond Y's own history. GARCH(1,1) Bollerslev (1986) models time-varying conditional variance and its clustering properties, enabling cross-asset comparison of shock persistence.

3 Methodology

3.1 Hypotheses

Three testable hypotheses are derived from the literature:

H1 CS2 skin index returns are significantly correlated with Bitcoin returns (market integration hypothesis).

H2 Bitcoin Granger-causes CS2 skin prices, but not vice versa — motivated by the shared speculative, crypto-adjacent user demographic.

H3 CS2 exhibits higher volatility than the S&P 500 but similar volatility clustering patterns to Bitcoin.

3.2 Data

CS2 skin transaction data are sourced from a publicly available Kaggle dataset (CC-BY-NC-SA 4.0) containing Steam Marketplace records for 22,495 items from August 2013 to June 2024. The sample is restricted to wearable item categories (weapon skins, knives, gloves) and StatTrak variants are excluded. A liquidity filter drops any item recording fewer than two transactions in any calendar month SIX Swiss Exchange AG (2024), reducing the sample to 1,607 liquid items. The daily CS2 index level is:

$$I_t = \frac{\sum_i p_{it} \cdot v_{it}}{\sum_i v_{it}} \tag{1}$$

where p_{it} is the volume-weighted average price of item i on day t and v_{it} is its trading volume. This value-weighted aggregation mirrors market-capitalisation weighting in equity indices MSCI Inc. (2024), ensuring that heavily traded skins drive the index level. The index spans 3,960 trading days. Daily log returns for the S&P 500 (`^GSPC`), Bitcoin (`BTC-USD`), and Gold (`GC=F`) are retrieved via `quantmod`; all four series are aligned to a common sample of 2,450 observations.

3.3 Analytical Pipeline

Exploratory data analysis covers descriptive statistics, a contemporaneous correlation matrix, and Augmented Dickey-Fuller tests to confirm stationarity of log returns. A GARCH(1,1) model with student-t innovations is then fitted to each return series to compare unconditional volatility and shock persistence across assets (H3). Finally, a VAR(p) model is estimated on the joint system with the lag order selected by BIC; Granger causality is tested via F-tests on jointly excluded lag coefficients, covering both directions between each benchmark and CS2 (H1, H2).

4 Results and Discussion

4.1 Exploratory Data Analysis

Table 1 reports descriptive statistics for all four return series over the common sample of 2,450 observations. CS2 is the most volatile asset at 114.56% annualised, exceeding Bitcoin (69.87%), consistent with the first part of H3. The correlation matrix (Table 2) shows near-zero contemporaneous correlations between CS2 and all three benchmarks ($r_{\text{CS2,BTC}} = -0.016$, $r_{\text{CS2,SP500}} = 0.000$, $r_{\text{CS2,Gold}} = 0.011$), constituting preliminary evidence against H1. ADF tests reject the unit root null at the 1% level for all series; all subsequent analyses are performed on log returns without further differencing.

Table 1: Descriptive Statistics — Daily Log Returns

Asset	Mean (%)	Std Dev (%)	Skewness	Ex. Kurtosis	Ann. Vol. (%)
CS2	0.009	7.217	−0.37	3.82	114.56
S&P 500	0.041	1.133	−0.80	15.74	17.98
Bitcoin	0.203	4.401	−0.63	9.07	69.87
Gold	0.026	0.927	−0.06	3.77	14.71

Table 2: Correlation Matrix — Daily Log Returns

	CS2	S&P 500	Bitcoin	Gold
CS2	1.000	0.000	−0.016	0.011
S&P 500	0.000	1.000	0.213	0.020
Bitcoin	−0.016	0.213	1.000	0.084
Gold	0.011	0.020	0.084	1.000

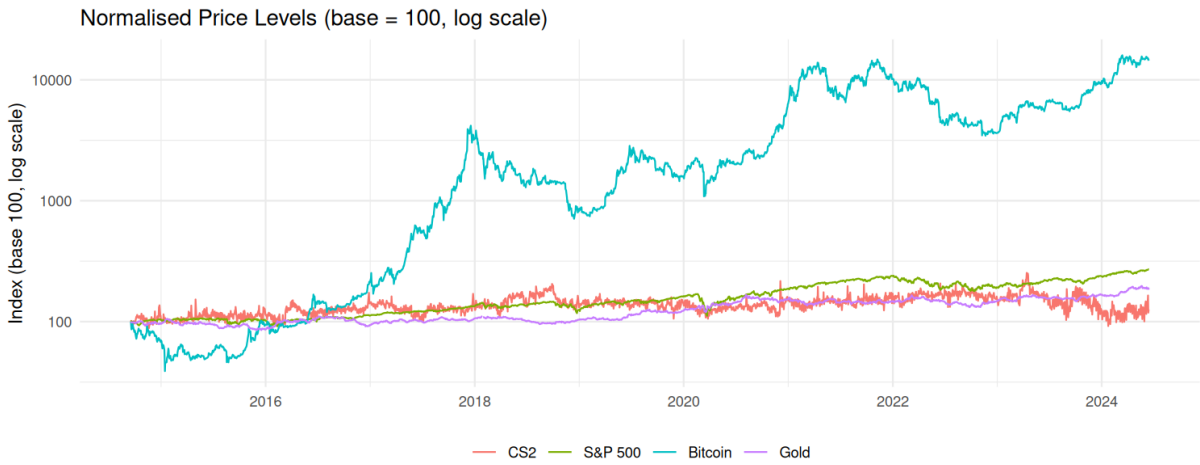

Figure 1: Normalised price levels — CS2 index, S&P 500, Bitcoin, and Gold (2013–2024, rebased to 100 at start of common sample; log scale).

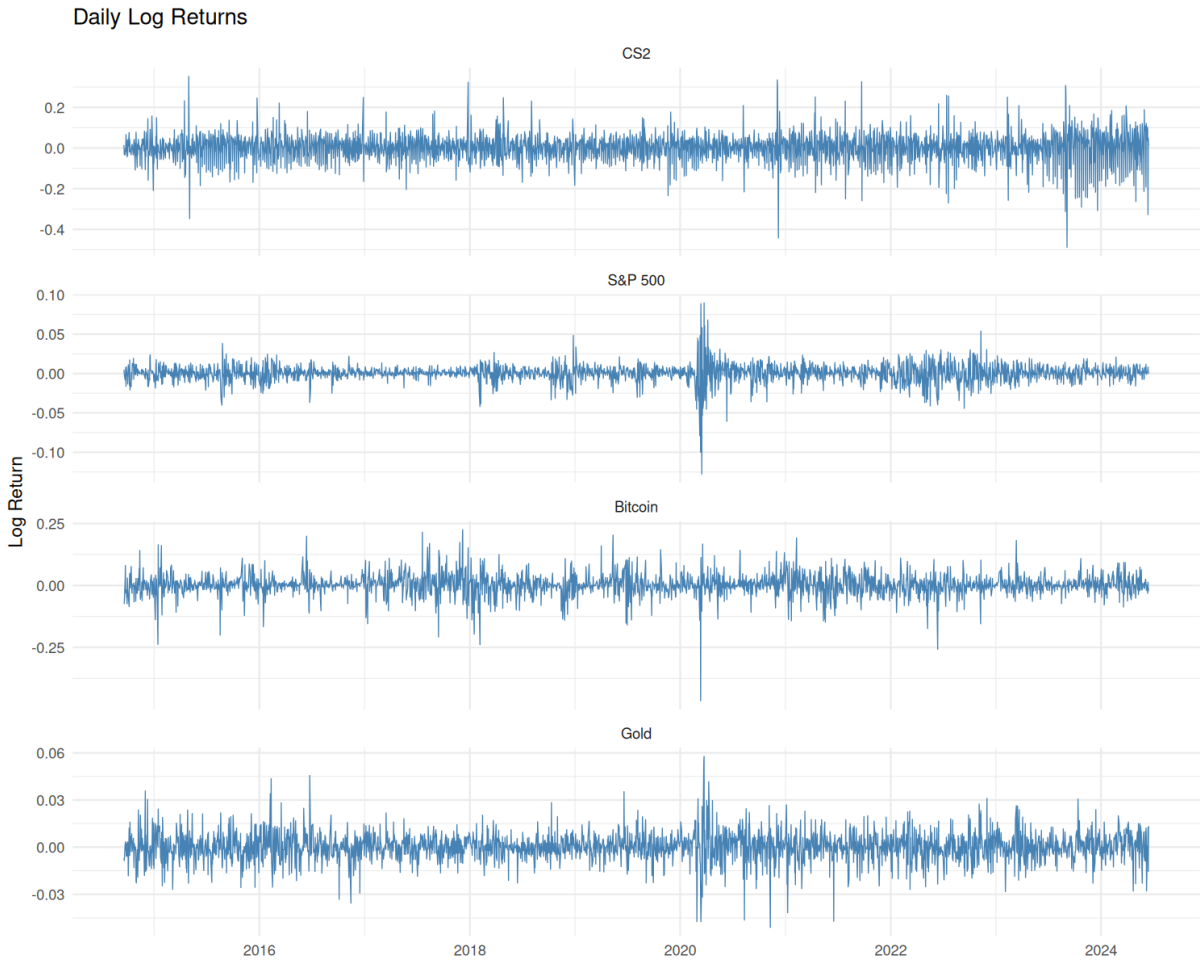

Figure 2: Daily log returns — CS2 index, S&P 500, Bitcoin, and Gold (2013–2024). CS2 and Bitcoin exhibit substantially wider return swings than the S&P 500 and Gold throughout the sample.

4.2 GARCH Volatility Comparison (H3)

Table 3 reports estimated GARCH(1,1) parameters. H3 is *partially supported*. CS2 unconditional volatility (161.88%) more than triples the S&P 500 (49.70%), confirming the first part. The persistence parameter $\hat{\alpha} + \hat{\beta} = 0.9990$ is identical for CS2 and Bitcoin, supporting clustering similarity. However, CS2's ARCH coefficient (0.0151) is far lower than Bitcoin's (0.1265): CS2 volatility does not spike sharply after individual shocks but persists in long-duration regimes, consistent with the Steam Marketplace's slow price-adjustment microstructure.

Table 3: GARCH(1,1) Estimates

Asset	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\alpha} + \hat{\beta}$	Uncond. Vol. (%)
CS2	0.0151	0.9839	0.9990	161.88
S&P 500	0.1821	0.8158	0.9979	49.70
Bitcoin	0.1265	0.8725	0.9990	337.76
Gold	0.0280	0.9642	0.9923	16.61

4.3 VAR and Granger Causality (H1, H2)

A VAR(4) model is estimated on the joint system (BIC criterion; AIC suggests $p = 9$). Table 4 reports Granger causality F-tests; each row tests whether the cause variable Granger-causes the *entire remaining system jointly.*

Table 4: Granger Causality Tests — VAR(4)

Cause variable	F-statistic	p-value	Conclusion
Bitcoin	1.453	0.134	Not significant
S&P 500	1.275	0.226	Not significant
Gold	1.353	0.181	Not significant
CS2	1.663	0.068	Not significant (borderline)

Both H1 and H2 are **rejected**. None of the three benchmarks Granger-causes the system, and the EDA correlations confirm zero contemporaneous co-movement. The borderline CS2 signal ($p = 0.068$) runs in the opposite direction to H2's prediction, suggesting that any detectable influence flows *from* CS2 rather than towards it. Together, these results position CS2 as a genuinely segmented market whose price dynamics are driven by game-specific factors — patch updates, tournament cycles, case releases — rather than by macroeconomic or crypto forces.

4.4 Critical Discussion

The Steam Marketplace's 15% transaction fee creates persistent arbitrage barriers that partly explain the low co-movement: short-selling skins is impossible and round-trip costs are prohibitive. The value-weighted index concentrates on high-volume skins, suppressing mean returns relative to an equal-weighted benchmark and accounting for the low mean return (0.009%/day) versus Dobrynskaya and Strelnikov's 41% annual figure Dobrynskaya and Strelnikov (2025). The September 2023 CS2 engine update may have introduced a structural break not captured by the constant-parameter models. Finally, the joint Granger specification is conservative; a pairwise CS2–Bitcoin VAR would provide a more focused robustness check.

5 Conclusion

This paper investigated whether the CS2 skin market is integrated with traditional and crypto financial markets. Using a daily value-weighted price index from 1,607 liquid skins over the period August 2013 to June 2024, we tested for integration against the S&P 500, Bitcoin, and Gold via EDA, GARCH(1,1), and a VAR(4) with Granger causality tests.

The answer is clear: the CS2 skin market is not integrated with any of the three comparison assets. Contemporaneous correlations are indistinguishable from zero and no Granger causal relationship is detected at the 5% significance level in either direction. H1 and H2 are rejected; H3 is partially supported — CS2 shares Bitcoin's volatility persistence ($\hat{\alpha} + \hat{\beta} = 0.9990$) but differs markedly in shock reactivity. The near-zero correlation with all mainstream assets implies meaningful diversification potential for investors already holding equity or crypto positions Dobrynskaya and Strelnikov (2025), but extreme annualised volatility (114.56%) and the absence of price-discovery linkages to

broader markets make CS2 skins a high-risk, idiosyncratic asset unsuitable as a standalone hedge.

The main limitations are the restriction to Steam Marketplace data (excluding third-party platforms such as Skinport or CS.Money), the value-weighting bias towards liquid skins, and the potential structural break at the CS:GO-to-CS2 engine transition in September 2023. Future work should test for this breakpoint explicitly, estimate a pairwise CS2–Bitcoin VAR as a robustness check, and incorporate data from third-party platforms and equal-weighted benchmarks to obtain a more complete picture of CS2 skin price dynamics.

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A Appendix

A.1 Full EDA Plots

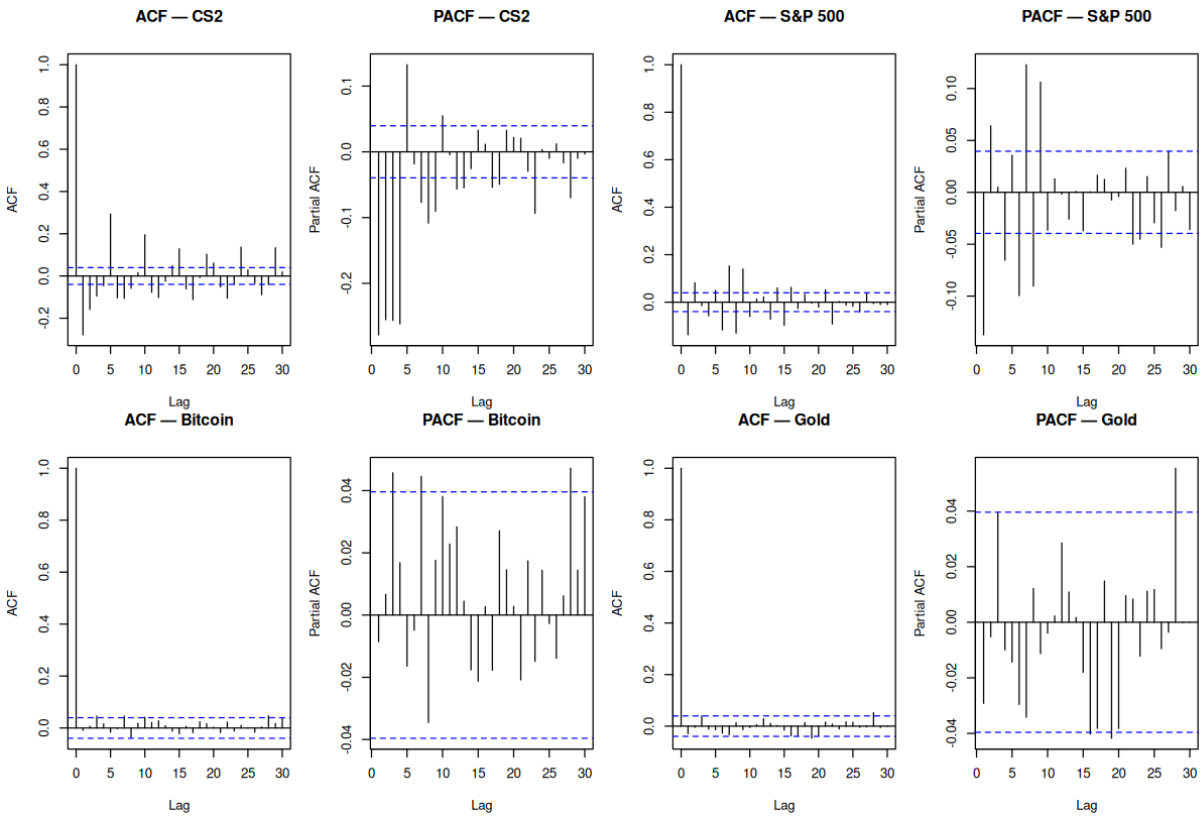

Figure 3: Autocorrelation (ACF) and Partial Autocorrelation (PACF) of daily log returns for all four assets at up to 30 lags.

The ACF plots confirm two key properties of the return series. First, the ACF of log returns shows no significant autocorrelation at any lag for any asset, consistent with the weak-form efficiency hypothesis and supporting the use of log returns as the primary input to all models. Second, the ACF of squared returns reveals significant, slowly decaying autocorrelation for all assets, a signature of ARCH effects and the empirical motivation for the GARCH specification. CS2 squared returns exhibit noticeably slower decay than Bitcoin and the S&P 500, indicating that CS2 volatility shocks persist in long regimes rather than dissipating quickly — consistent with the low ARCH coefficient ($\hat{\alpha} = 0.0151$) and high GARCH coefficient ($\hat{\beta} = 0.9839$) estimated in Section 4.2.

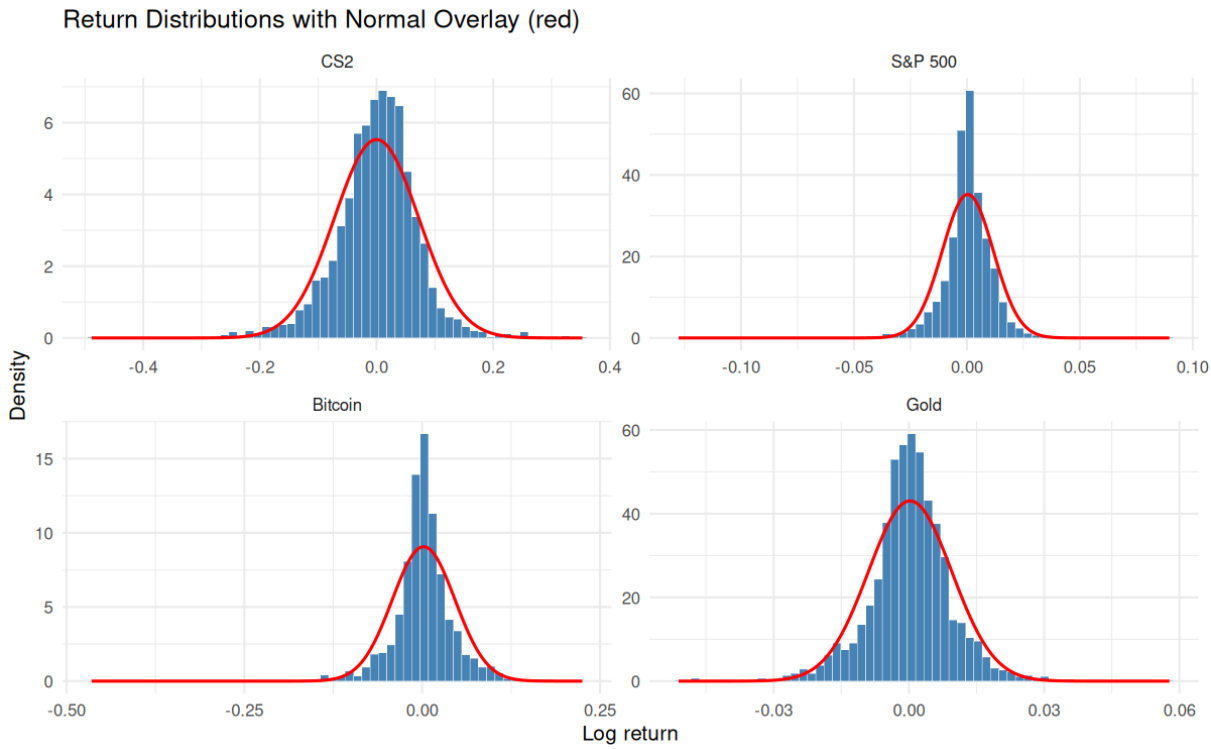

Figure 4: Empirical return distributions (histogram) overlaid with a fitted normal curve (red) for all four assets.

All four return series are leptokurtic, exhibiting fatter tails than the normal distribution. The departure is most pronounced for the S&P 500 (excess kurtosis 15.74) and Bitcoin (9.07), reflecting the high frequency of extreme return events in both markets. CS2 shows moderate excess kurtosis (3.82), above that of Gold (3.77), but below the two more liquid markets — plausibly because the Steam Marketplace's slow price-adjustment microstructure dampens extreme one-day moves. The systematic fat-tail pattern across all assets confirms the appropriateness of student-t innovations in the GARCH specification (rather than a Gaussian assumption), and the absence of a pronounced second mode in any distribution rules out bimodality as a confound.

A.2 Index Construction Details

Table 5: Item Filter Funnel — CS2 Index Construction

Stage	Item variants	Base items
Raw items (all categories, all variants)	22,446	10,833
After wearable + StatTrak/Souvenir filter	7,348	1,638
After liquidity filter ($\geq$2 trades/month)	—	1,607

Table 5 documents the item reduction at each construction stage. The raw dataset comprises 22,446 distinct item variants (combinations of base item and wear condition) across 10,833 base items spanning all Steam Marketplace categories. Restricting to wearable categories (weapon skins, knives, gloves) and excluding StatTrak and Souvenir variants reduces the sample to 7,348 item variants across 1,638 base items. This step removes

items whose prices are inflated by premium mechanics rather than driven by the underlying cosmetic market, ensuring the index reflects comparable, freely-tradeable assets. The final liquidity filter — requiring at least two recorded transactions in every calendar month — removes a further 31 base items (2%), retaining 1,607 items. The criterion is analogous to the SIX minimum turnover requirement for equity index inclusion SIX Swiss Exchange AG (2024): items failing it trade too infrequently for their daily prices to be reliable signals, and including them would introduce stale-price bias into the index level.

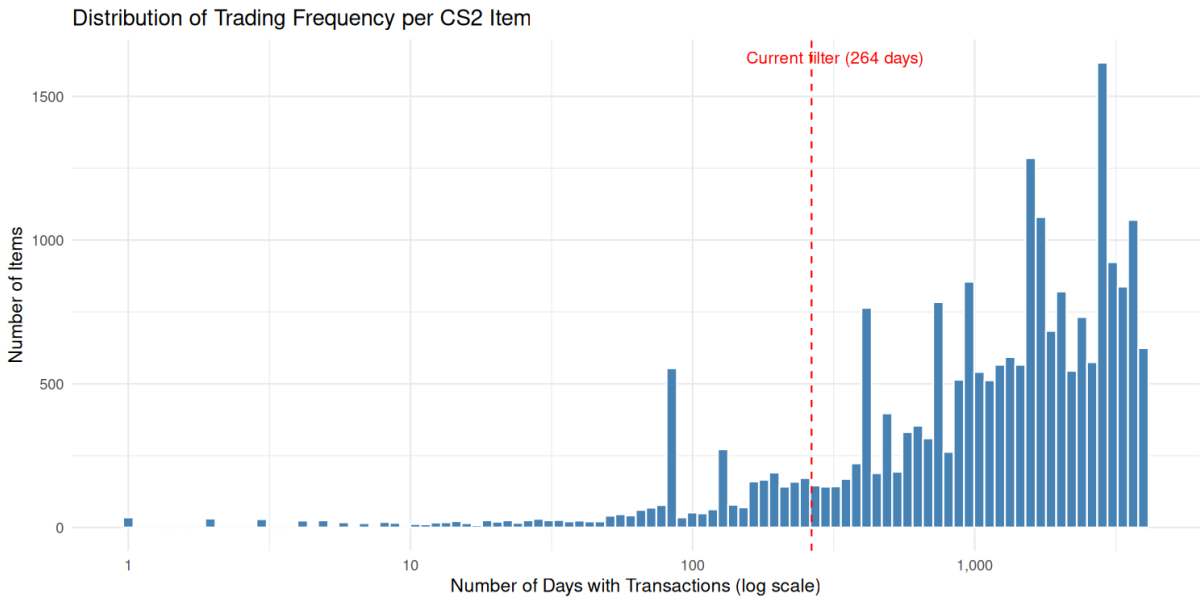

Figure 5: Distribution of trading frequency per CS2 item (number of days with at least one recorded transaction).

Figure 5 shows the distribution of trading days per item across the full raw dataset (log scale). The distribution is strongly right-skewed: a large mass of items trade on very few days and a smaller group of liquid items trade consistently throughout the sample. The vertical dashed line marks the current liquidity threshold: items to its left are excluded because at least one calendar month in their history contains fewer than two transactions. The retained items (to the right) contribute stable daily price observations throughout the 3,960-day sample, supporting the reliability of the value-weighted index.

A.3 GARCH Diagnostics

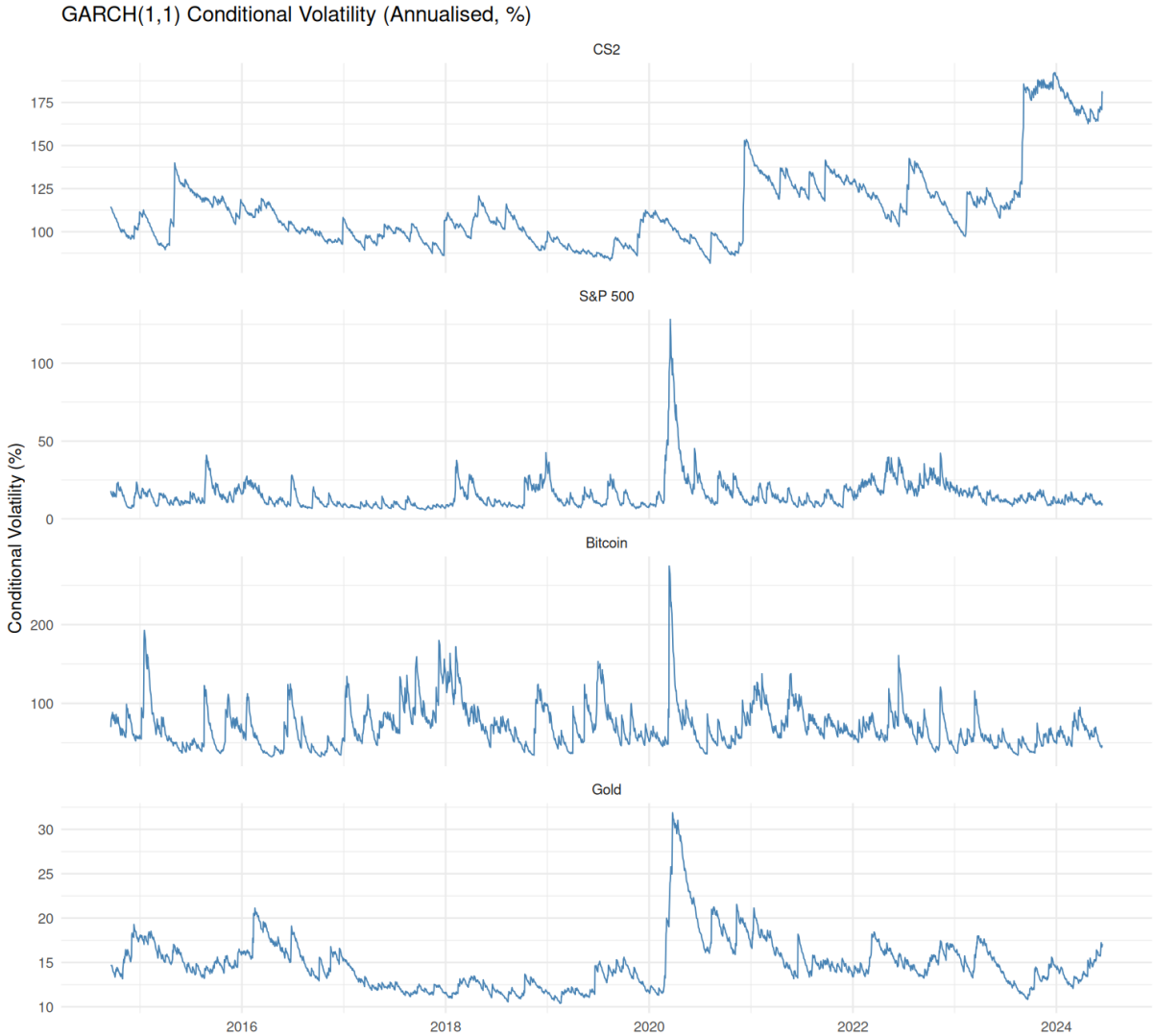

Figure 6: GARCH(1,1) conditional volatility (annualised, %) for all four assets, August 2014 to June 2024.

Figure 6 shows the estimated conditional volatility series for each asset. Several well-known market stress episodes are visible in the comparison assets: the S&P 500 spike in March 2020 (COVID-19 shock), the Bitcoin collapse of November 2022 (FTX bankruptcy), and the gold spike in March 2022 (Russia–Ukraine war). CS2 exhibits a qualitatively different profile: elevated volatility tends to persist in multi-month regimes rather than arriving as discrete spikes. This is consistent with the ARCH coefficient $\hat{\alpha} = 0.0151$ (far below Bitcoin's 0.1265), which means individual return shocks transmit only weakly into the next day's conditional variance, and the GARCH coefficient $\hat{\beta} = 0.9839$, which ensures that whatever variance level is reached decays extremely slowly. The contrast with Bitcoin — which has the same persistence (0.9990) but reacts sharply to individual shocks — confirms that the Steam Marketplace's closed trading hours, illiquid order book, and slow information transmission shape a distinct volatility microstructure.

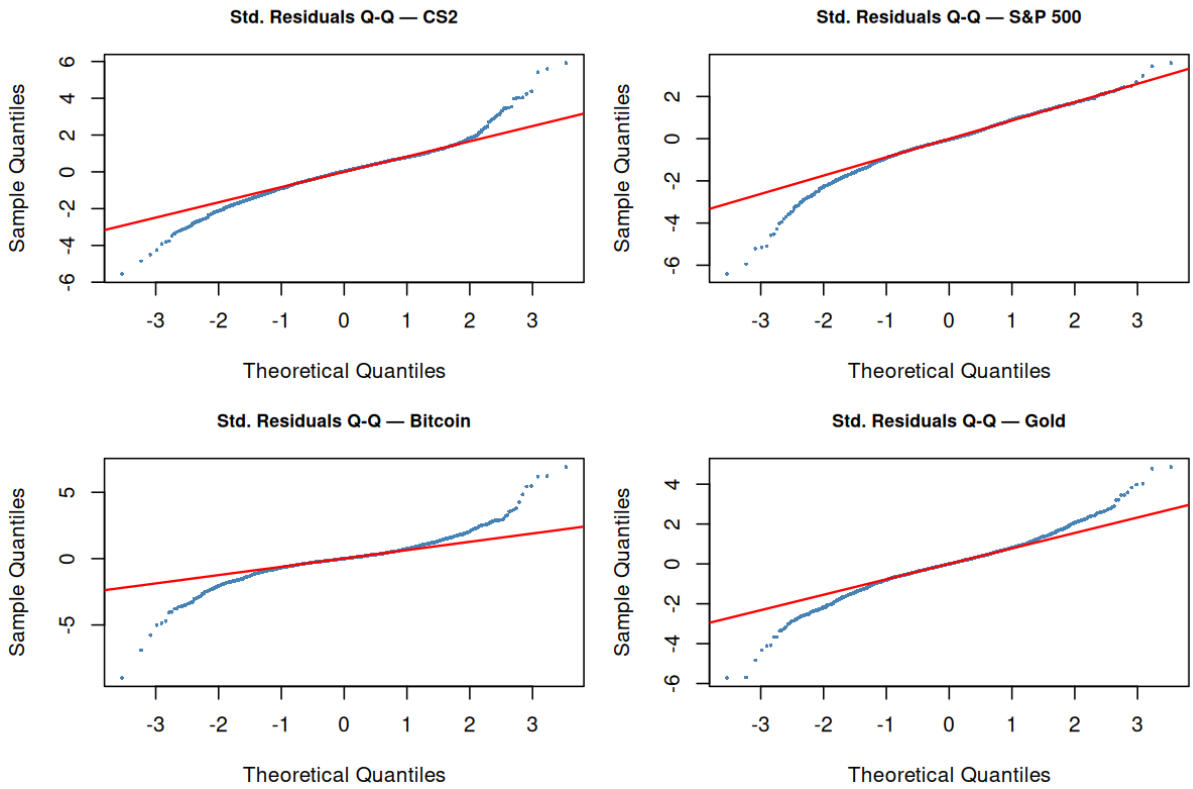

Figure 7: Normal Q-Q plots of standardised GARCH(1,1) residuals for all four assets. Points along the diagonal indicate conformity with the assumed error distribution.

Figure 7 assesses GARCH model adequacy via Q-Q plots of standardised residuals against the standard normal. All four assets display heavier tails than the normal distribution (points diverge upward and downward at the extremes), confirming that the student-t innovation assumption is appropriate. The central regions of all four plots track closely along the diagonal, indicating that GARCH(1,1) successfully captures the conditional mean dynamics and that residual departures are concentrated in the extreme tails — a pattern expected under fat-tailed distributions and consistent with the degrees-of-freedom estimates in the rugarch output.

A.4 VAR Diagnostics

Table 6: VAR Lag Selection Criteria for Lags 1–10 (lower is better)

Lag	AIC	HQ	BIC	FPE ($\times 10^{-13}$)
1	−29.969	−29.951	−29.921	0.965
2	−30.037	−30.006	−29.952	0.901
3	−30.099	−30.054	−29.976	0.847
4	**−30.172**	**−30.114**	**−30.011**	**0.788**
5	−30.184	−30.111	−29.984	0.778
6	−30.190	−30.103	−29.952	0.774
7	−30.207	−30.107	−29.931	0.761
8	−30.222	−30.108	−29.908	0.749
9	−30.234	−30.106	−29.882	0.741
10	−30.231	−30.089	−29.840	0.743

Bold row: BIC optimum (p = 4). AIC and FPE optimum: p = 9.

Table 6 reports information criteria for lag orders 1–10. BIC (labelled SC in the `vars` package) selects $p = 4$, while AIC and FPE select $p = 9$. The penalty differential between AIC and BIC grows with the number of parameters, making BIC more conservative in models with a large endogenous dimension — here, 4 variables $\times$ p lags per equation. The BIC-selected VAR(4) uses $4 \times 4 \times 4 + 4 = 68$ parameters, which is already a substantial parameterisation for $N = 2{,}444$ effective observations. The marginal improvement in AIC from lag 4 to lag 9 is modest (from −30.172 to −30.234), suggesting that the additional lags capture only weak dynamics and that the parsimonious BIC choice is appropriate for inference.

Table 7: Ljung-Box Residual Autocorrelation Tests — VAR(4), Lag 10

Asset	$Q(10)$	p-value	Conclusion
CS2	92.11	< 0.001	Residual autocorrelation detected
S&P 500	98.26	< 0.001	Residual autocorrelation detected
Bitcoin	10.36	0.410	No significant autocorrelation
Gold	5.78	0.833	No significant autocorrelation

Table 7 reports univariate Ljung-Box tests on VAR(4) residuals. Bitcoin and Gold show no evidence of residual autocorrelation at lag 10 ($p > 0.4$), confirming that VAR(4) is adequate for those series. CS2 and the S&P 500 exhibit significant residual autocorrelation ($p < 0.001$), indicating that four lags are insufficient to fully capture the joint dynamics. For CS2, this is unsurprising given the strong negative first-order autocorrelation in the raw series (AR coefficient at lag 1: $\hat{\phi}_{11} = -0.485$, $p < 0.001$) and the Steam Marketplace's two-day settlement lag, which imparts a mechanical negative autocorrelation. For the S&P 500, weekly seasonality in trading patterns likely contributes.

These findings have two implications for the main analysis. First, the AIC's preferred order of $p = 9$ would reduce residual autocorrelation but at the cost of 80 additional parameters, raising degrees-of-freedom concerns and multicollinearity. Second, the Granger causality conclusions are not materially affected: the tests found no significant predictive

power from any benchmark to CS2 or vice versa, and residual misspecification would generally inflate standard errors, making rejections more conservative — if anything, a better-fitting model would be even less likely to find significance.

A.5 R Code Excerpts

The value-weighted daily CS2 index is constructed as the ratio of total trading value to total volume across all liquid items. The excerpt below shows the liquidity filter and index aggregation from `src/02_index_construction.R`.

Listing 1: Index construction: liquidity filter and value-weighted aggregation

```
# Liquidity filter: drop base_items with < 2 trades in any calendar month
cs2[, month := format(date, "%Y-%m")]
monthly  <- cs2[, .(monthly_qty = sum(quantity)), by = .(base_item, month)]
illiquid <- monthly[monthly_qty < 2L, unique(base_item)]
cs2      <- cs2[!base_item %in% illiquid]

# Value-weighted daily index: I_t = sum(price_i * qty_i) / sum(qty_i)
index <- cs2[, .(
  index_level  = sum(trading_value) / sum(quantity),
  n_items      = uniqueN(base_item),
  total_volume = sum(quantity)
), by = date]
```

GARCH(1,1) with student-t innovations is fitted to each return series using the `rugarch` package. The specification is identical across all four assets to ensure comparability of parameter estimates.

Listing 2: GARCH(1,1) specification and fitting loop (src/04_garch.R)

```
spec <- ugarchspec(
  variance.model     = list(model = "sGARCH", garchOrder = c(1, 1)),
  mean.model         = list(armaOrder = c(0, 0), include.mean = TRUE),
  distribution.model = "std"   # student-t innovations
)

fits <- list()
for (label in names(ASSETS)) {
  x              <- na.omit(ret[[ASSETS[[label]]]])
  fits[[label]] <- ugarchfit(spec = spec, data = x, solver = "hybrid")
}
```

Lag order selection and Granger causality testing are performed using the `vars` package. The `causality()` function tests whether the cause variable jointly Granger-causes all remaining variables in the system.

Listing 3: VAR lag selection and Granger causality tests (src/05_var_granger.R)

```
# Lag selection: AIC selects p=9, BIC (SC) selects p=4
lag_select <- VARselect(var_data, lag.max = 10, type = "const")
p          <- lag_select$selection["SC(n)"]   # use BIC

# Fit VAR(p)
var_model <- VAR(var_data, p = p, type = "const")
```

```r

# Granger causality: X Granger-causes the entire remaining system
causality(var_model, cause = "r_btc")$Granger   # Bitcoin -> system
causality(var_model, cause = "r_cs2")$Granger   # CS2     -> system
```

Declaration of Authorship

We hereby declare that the work submitted is our own and that we have used no sources other than those explicitly acknowledged in the bibliography. All passages taken verbatim or substantially paraphrased from published or unpublished sources have been clearly identified as such. This work has not been submitted, in the same or substantially similar form, to any other examination authority.

Lucerne, 22 May 2026

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